

ProspectSearchAgent

ProspectSearchAgent is a Python tool that aggregates company and contact data from multiple sources, enriches it with tech stack and hiring signals, and outputs a clean, deduplicated list of prospects based on your Ideal Customer Profile (ICP).

The tool works with real APIs if keys are provided, but can also use mock data for demo purposes.

Features

- Fetch companies and contacts from **Apollo.io** (or fallback mock data).
- Fetch funding & financial data from **Crunchbase** (or mock fallback).
- Detect technology stack using **BuiltWith** API.
- Fetch hiring/job signals using **SerpAPI**.
- Merge and deduplicate results from multiple sources.
- Configurable **ICP** in YAML/JSON.
- Outputs results in **JSON** format.
- Async-ready for faster API calls.

Recommended Tools / APIs

Function	Tool / API	Free Tier	Notes
Company & Contact Search	Apollo.io API	✅ Free developer account (100 credits)	Rich company + contact data
Funding / Financials	Crunchbase API	✅ Free developer API	Fetch funding rounds
Tech Stack Info	BuiltWith API	✅ Free trial (10 lookups)	Detect site technologies

Hiring / Job Signals	SerpAPI	✓ Free 100 searches/month	Scrape Google Jobs / LinkedIn
Email Verification (optional)	Hunter.io API	✓ Free 25 searches/month	Verify emails

Project Structure

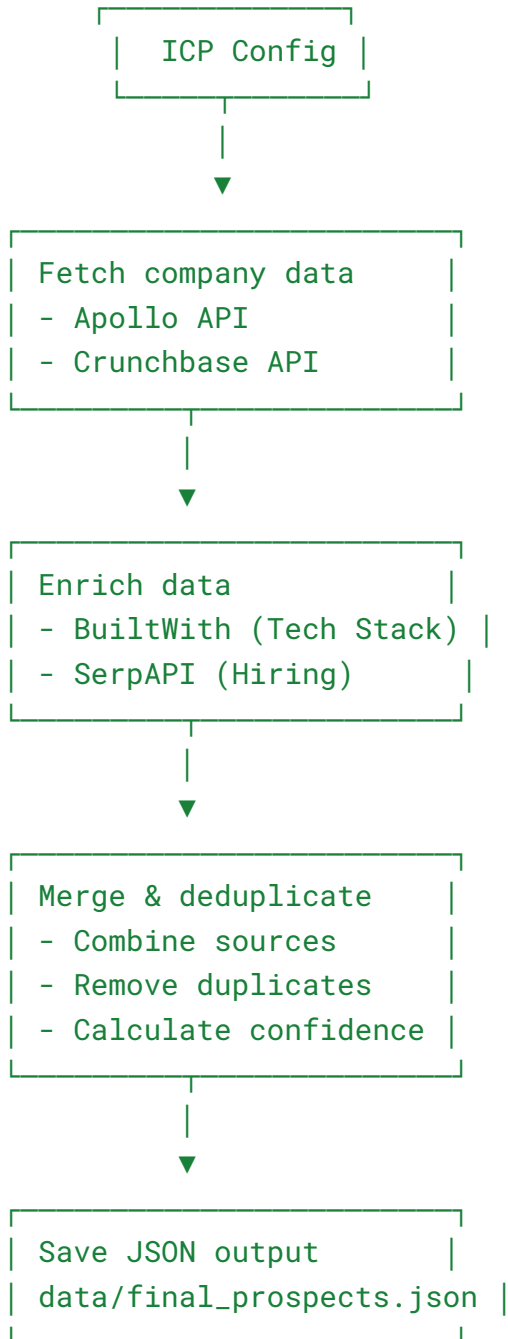
```

ProspectSearchAgent/
├── data/                # Output JSON
├── config/              # ICP configuration
│   └── icp_config.yaml
├── src/
│   └── main.py          # Orchestrates async API calls & merges
data
├── apollo_api.py        # Apollo API / mock fallback
├── crunchbase_api.py    # Crunchbase API / mock fallback
├── techstack_api.py     # BuiltWith API / mock fallback
├── serpapi_signals.py   # SerpAPI / hiring signals
├── icp_parser.py        # Load ICP configuration
├── merge_results.py     # Merge and deduplicate results
├── .env                 # API keys
├── requirements.txt     # Python dependencies
└── README.md            # Project overview

```

How it Works

Workflow diagram:



Step-by-step explanation:

1. **ICP Config** – The user defines their Ideal Customer Profile in YAML/JSON format.
2. **Data Fetching** – The system queries Apollo and Crunchbase APIs (or uses mock data if keys are missing).
3. **Data Enrichment** – Each company is enriched with:
 - Technology stack from BuiltWith API
 - Hiring signals from SerpAPI
4. **Merge & Deduplicate** – Data from multiple sources is combined, duplicates removed, confidence calculated.
5. **Output** – Final cleaned data is saved in `data/final_prospects.json`.

Setup

1. Clone the repository:

```
git clone https://github.com/yourusername/ProspectSearchAgent.git
cd ProspectSearchAgent
```

2. Create a virtual environment:

```
python -m venv venv
source venv/bin/activate      # Linux / macOS
venv\Scripts\activate        # Windows
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Add your API keys to `.env`:

```
APOLLO_API_KEY=your_apollo_key
CRUNCHBASE_API_KEY=your_crunchbase_key
BUILTWITH_API_KEY=your_builtwith_key
SERPAPI_KEY=your_serpapi_key
```

5. Configure ICP in `config/icp_config.yaml`. Example:

```
revenue_min: 20000000
revenue_max: 200000000
industry:
  - "B2B Software"
  - "FinTech"
geography:
  - "USA"
employee_count_min: 100
keywords:
  - "AI"
  - "data analytics"
  - "automation"
  - "machine learning"
signals:
  funding: true
  hiring_data_roles: true
  tech_stack:
    - "Snowflake"
    - "AWS"
```

Usage

Run the main script:

```
python src/main.py
```

The final merged prospects will be saved in:

```
data/final_prospects.json
```

Notes

- Works with **mock data** if APIs are not available.
- Supports **async fetching** for faster performance.